

CHAPTER 1

GENERAL

1.1 SCOPE

This standard provides minimum loads, hazard levels, associated criteria, and intended performance goals for buildings, other structures, and their nonstructural components that are subject to building code requirements. The loads, load combinations, and associated criteria provided herein are to be used with design strengths or allowable stress limits contained in design specifications for conventional structural materials. Used together, they are deemed capable of providing the intended performance levels for which the provisions of this standard have been developed. Procedures for applying alternative means to demonstrate acceptable performance are also described.

1.2 DEFINITIONS AND SYMBOLS

1.2.1 Definitions. The following definitions apply to the provisions of the entire standard.

ALLOWABLE STRESS DESIGN: A method of proportioning structural members such that elastically computed stresses produced in the members by nominal loads do not exceed specified allowable stresses (also called “working stress design”).

AUTHORITY HAVING JURISDICTION: The organization, political subdivision, office, or individual charged with the responsibility of administering and enforcing the provisions of this standard.

BUILDINGS: Structures, usually enclosed by walls and a roof, constructed to provide support or shelter for an intended occupancy.

DESIGN STRENGTH: The product of the nominal strength and a resistance factor.

DESIGNATED NONSTRUCTURAL SYSTEM: A non-structural component or system that is essential to the intended function of a Risk Category IV structure or that is essential to Life Safety in structures assigned to other Risk Categories.

ESSENTIAL FACILITIES: Buildings and other structures that are intended to remain operational in the event of extreme environmental loading from flood, wind, snow, or earthquakes.

FACTORED LOAD: The product of the nominal load and a load factor.

HIGHLY TOXIC SUBSTANCE: As defined in 29 CFR 1910.1200, Appendix A, with Amendments as of February 1, 2000.

IMPORTANCE FACTOR: A factor that accounts for the degree of risk to human life, health, and welfare associated with damage to property or loss of use or functionality.

LIMIT STATE: A condition beyond which a structure or member becomes unfit for service and is judged either to be no longer useful for its intended function (serviceability limit state) or to be unsafe (strength limit state).

LOAD EFFECTS: Forces and deformations produced in structural members by the applied loads.

LOAD FACTOR: A factor that accounts for deviations of the actual load from the nominal load, for uncertainties in the analysis that transform the load into a load effect, and for the probability that more than one extreme load will occur simultaneously.

LOADS: Forces or other actions that result from the weight of all building materials, occupants and their possessions, environmental effects, differential movement, and restrained dimensional changes. Permanent loads are loads in which variations over time are rare or of small magnitude. All other loads are variable loads (see also “nominal loads”).

NOMINAL LOADS: The magnitudes of the loads specified in this standard for dead, live, soil, wind, snow, rain, flood, and earthquake loads.

NOMINAL STRENGTH: The capacity of a structure or member to resist the effects of loads, as determined by computations using specified material strengths and dimensions and formulas derived from accepted principles of structural mechanics or by field tests or laboratory tests of scaled models, allowing for modeling effects and differences between laboratory and field conditions.

OCCUPANCY: The purpose for which a building or other structure, or part thereof, is used or intended to be used.

OTHER STRUCTURES: Structures, other than buildings, for which loads are specified in this standard.

P-DELTA EFFECT: The second-order effect on shears and moments of frame members induced by axial loads on a laterally displaced building frame.

PERFORMANCE-BASED PROCEDURES: An alternative to the prescriptive procedures in this standard characterized by project-specific engineering analysis, optionally supplemented by limited testing, to determine the computed reliability of an individual building or structure.

RESISTANCE FACTOR: A factor that accounts for deviations of the actual strength from the nominal strength and the manner and consequences of failure (also called “strength reduction factor”).

RISK CATEGORY: A categorization of buildings and other structures for determination of flood, snow, ice, and earthquake loads based on the risk associated with unacceptable performance. See Table 1.5-1.

SERVICE LOADS: Loads imparted on a building or other structure because of (1) self-weight and superimposed dead load, (2) live loads assumed to be present during normal occupancy or use of the building or other structure, (3) environmental loads that are expected to occur during the defined service life of a building or other structure, and (4) self-straining forces and effects. Service live loads and environmental loads for a

particular limit state are permitted to be less than the design loads specified in the standard. Service loads shall be identified for each serviceability state being investigated.

STRENGTH DESIGN: A method of proportioning structural members such that the computed forces produced in the members by the factored loads do not exceed the member design strength (also called “load and resistance factor design”).

TEMPORARY FACILITIES: Buildings or other structures that are to be in service for a limited time and have a limited exposure period for environmental loadings.

TOXIC SUBSTANCE: As defined in 29 CFR 1910.1200, Appendix A, with Amendments as of February 1, 2000.

1.2.2 Symbols.

D = Dead load.

F_x = A minimum design lateral force applied to level x of the structure and used for purposes of evaluating structural integrity in accordance with Section 1.4.2.

L = Live load.

L_r = Roof live load.

N = Notional load for structural integrity.

R = Rain load.

S = Snow load.

W_x = The portion of the total dead load of the structure, D , located or assigned to level x .

1.3 BASIC REQUIREMENTS

1.3.1 Strength and Stiffness. Buildings and other structures, and all parts thereof, shall be designed and constructed with adequate strength and stiffness to provide structural stability, protect nonstructural components and systems, and meet the serviceability requirements of Section 1.3.2.

Acceptable strength shall be demonstrated using one or more of the following procedures:

- the strength procedures of Section 1.3.1.1,
- the allowable stress procedures of Section 1.3.1.2, or
- subject to the approval of the Authority Having Jurisdiction for individual projects, the performance-based procedures of Section 1.3.1.3.

It shall be permitted to use alternative procedures for different parts of a structure and for different load combinations, subject to the limitations of Chapter 2. Where resistance to extraordinary events is considered, the procedures of Section 2.5 shall be used.

1.3.1.1 Strength Procedures. Structural and nonstructural components and their connections shall have adequate strength to resist the applicable load combinations of Section 2.3 of this standard without exceeding the applicable strength limit states for the materials of construction.

1.3.1.2 Allowable Stress Procedures. Structural and nonstructural components and their connections shall have adequate strength to resist the applicable load combinations of Section 2.4 of this standard without exceeding the applicable allowable stresses for the materials of construction.

1.3.1.3 Performance-Based Procedures. Structural and nonstructural components and their connections designed with performance-based procedures shall be demonstrated by analysis in accordance with Section 2.3.6 or by analysis procedures supplemented by testing to provide a reliability that is generally consistent with the target reliabilities stipulated in this section. Structural and nonstructural components subjected to dead, live, environmental, and other loads except earthquake, tsunami, flood, and loads from extraordinary events shall be based on the target reliabilities in Table 1.3-1. Structural systems subjected to earthquake shall be based on the target reliabilities in Tables 1.3-2 and 1.3-3. The analysis procedures used shall account for uncertainties in loading and resistance.

Testing methods in Sections 1.3.1.3.2 shall only be applied to individual projects and shall not be applied to development of values of material resistance for general use in structural systems.

Structures and nonstructural components shall meet the serviceability and functionality requirements of Sections 1.3.2 and 1.3.3.

Performance-based design provisions for structures subjected to tsunami shall conform to the requirements of Chapter 6.

1.3.1.3.1 Analysis. Analysis shall use rational methods based on accepted principles of engineering mechanics and shall consider all significant sources of deformation and resistance. Assumptions of stiffness, strength, damping, and other properties of components and connections incorporated in the analysis shall be based on approved test data or referenced standards.

1.3.1.3.2 Testing. Testing used to substantiate the performance capability of structural and nonstructural components and their connections under load shall accurately represent the materials, configuration, construction, loading intensity, and boundary conditions anticipated in the structure. Where an approved industry standard or practice that governs the testing of similar components exists, the test program and determination of design

Table 1.3-1 Target Reliability (Annual Probability of Failure, P_F) and Associated Reliability Indices (β)¹ for Load Conditions That Do Not Include Earthquake, Tsunami, or Extraordinary Events²

Basis	Risk Category			
	I	II	III	IV
Failure that is not sudden and does not lead to widespread progression of damage	$P_F = 1.25 \times 10^{-4}/\text{yr}$ $\beta = 2.5$	$P_F = 3.0 \times 10^{-5}/\text{yr}$ $\beta = 3.0$	$P_F = 1.25 \times 10^{-5}/\text{yr}$ $\beta = 3.25$	$P_F = 5.0 \times 10^{-6}/\text{yr}$ $\beta = 3.5$
Failure that is either sudden or leads to widespread progression of damage	$P_F = 3.0 \times 10^{-5}/\text{yr}$ $\beta = 3.0$	$P_F = 5.0 \times 10^{-6}/\text{yr}$ $\beta = 3.5$	$P_F = 2.0 \times 10^{-6}/\text{yr}$ $\beta = 3.75$	$P_F = 7.0 \times 10^{-7}/\text{yr}$ $\beta = 4.0$
Failure that is sudden and results in widespread progression of damage	$P_F = 5.0 \times 10^{-6}/\text{yr}$ $\beta = 3.5$	$P_F = 7.0 \times 10^{-7}/\text{yr}$ $\beta = 4.0$	$P_F = 2.5 \times 10^{-7}/\text{yr}$ $\beta = 4.25$	$P_F = 1.0 \times 10^{-7}/\text{yr}$ $\beta = 4.5$

¹The target reliability indices are provided for a 50-year reference period, and the probabilities of failure have been annualized. The equations presented in Section 2.3.6 are based on reliability indices for 50 years because the load combination requirements in Section 2.3.2 are based on the maximum loads for the 50-year reference period.

²Commentary to Section 2.5 includes references to publications that describe the historic development of these target reliabilities.

Table 1.3-2 Target Reliability (Conditional Probability of Failure) for Structural Stability Caused by Earthquake

Risk Category	Conditional Probability of Failure Caused by the MCE_R Shaking Hazard (%)
I & II	10
III	5
IV	2.5

Table 1.3-3 Target Reliability (Conditional Probability of Failure) for Ordinary Noncritical Structural Members Caused by Earthquake

Risk Category	Conditional Probability of Component or Anchorage Failure Caused by the MCE_R Shaking Hazard (%)
I & II	25
III	15
IV	9

values from the test program shall be in accordance with those industry standards and practices. Where such standards or practices do not exist, specimens shall be constructed to a scale similar to that of the intended application unless it can be demonstrated that scale effects are not significant to the indicated performance. Evaluation of test results shall be made on the basis of the values obtained from not less than three tests, provided that the deviation of any value obtained from any single test does not vary from the average value for all tests by more than 15%. If such deviation from the average value for any test exceeds 15%, then additional tests shall be performed until the deviation of any test from the average value does not exceed 15% or a minimum of six tests have been performed. No test shall be eliminated unless a rationale for its exclusion is given. Test reports shall document the location, the time and date of the test, the characteristics of the tested specimen, the laboratory facilities, the test configuration, the applied loading and deformation under load, and the occurrence of any damage sustained by the specimen, together with the loading and deformation at which such damage occurred.

1.3.1.3.3 Documentation. The procedures used to demonstrate compliance with this section and the results of analysis and testing shall be documented in one or more reports submitted to the Authority Having Jurisdiction and to an independent peer review.

1.3.1.3.4 Peer Review. The procedures and results of analysis, testing, and calculation used to demonstrate compliance with the requirements of this section shall be subject to an independent peer review approved by the Authority Having Jurisdiction. The peer review shall comprise one or more persons having the necessary expertise and knowledge to evaluate compliance, including knowledge of the expected performance, the structural and component behavior, the particular loads considered, structural analysis of the type performed, the materials of construction, and laboratory testing of elements and components to determine structural resistance and performance characteristics. The review shall include assumptions, criteria, procedures, calculations, analytical models, test setup, test data, final drawings, and reports. Upon satisfactory completion, the peer reviewers

shall submit a letter to the Authority Having Jurisdiction indicating the scope of their review and their findings.

1.3.2 Serviceability. Structural systems, and members thereof, shall be designed under service loads to have adequate stiffness to limit deflections, lateral drift, vibration, or any other deformations that adversely affect the intended use and performance of buildings and other structures based on requirements set forth in the applicable codes and standards, or as specified in the project design criteria.

1.3.3 Functionality. Structural systems and members and connections thereof assigned to Risk Category IV shall be designed with reasonable probability to have adequate structural strength and stiffness to limit deflections, lateral drift, or other deformations such that their behavior would not prevent function of the facility immediately following any of the design level environmental hazard events specified in this standard. Designated nonstructural systems and their attachment to the structure shall be designed with sufficient strength and stiffness such that their behavior would not prevent function immediately following any of the design level environmental hazard events specified in this standard. Components of designated nonstructural systems shall be designed, qualified, or otherwise protected such that they shall be demonstrated capable of performing their critical function after the facility is subjected to any of the design level environmental hazards specified in this standard.

The provisions in Section 1.3.1.1 and Section 1.3.1.2 in this standard are deemed to comply with the requirements of this section.

1.3.4 Self-Straining Forces and Effects. Provision shall be made for anticipated self-straining forces and effects arising from differential settlements of foundations and from restrained dimensional changes caused by temperature, moisture, shrinkage, creep, and similar effects.

1.3.5 Analysis. Load effects on individual structural members shall be determined by methods of structural analysis that take into account equilibrium, general stability, geometric compatibility, and both short- and long-term material properties. Members that tend to accumulate residual deformations under repeated service loads shall have included in their analysis the added eccentricities expected to occur during their service life.

1.3.6 Counteracting Structural Actions. All structural members and systems, and all components and cladding in a building or other structure, shall be designed to resist forces caused by earthquake and wind, with consideration of overturning, sliding, and uplift, and continuous load paths shall be provided for transmitting these forces to the foundation. Where sliding is used to isolate the elements, the effects of friction between sliding elements shall be included as a force. Where all or a portion of the resistance to these forces is provided by dead load, the dead load shall be taken as the minimum dead load likely to be in place during the event causing the considered forces. Consideration shall be given to the effects of vertical and horizontal deflections resulting from such forces.

1.3.7 Fire Resistance. Structural fire resistance shall be provided in accordance with the requirements specified in the applicable building code. As an alternative, the performance-based design procedures in Appendix E are permitted, where approved.

1.4 GENERAL STRUCTURAL INTEGRITY

All structures shall be provided with a continuous load path in accordance with the requirements of Section 1.4.1 and shall have

a complete lateral force-resisting system with adequate strength to resist the forces indicated in Section 1.4.2. All members of the structural system shall be connected to their supporting members in accordance with Section 1.4.3. Structural walls shall be anchored to diaphragms and supports in accordance with Section 1.4.4. The effects on the structure and its components caused by the forces stipulated in this section shall be taken as the notional load, N , and combined with the effects of other loads in accordance with the load combinations of Section 2.6. Where material resistance depends on load duration, notional loads are permitted to be taken as having a duration of 10 minutes. Structures designed in conformance with the requirements of this standard for Seismic Design Categories B, C, D, E, or F shall be deemed to comply with the requirements of Sections 1.4.2, 1.4.3, and 1.4.4.

1.4.1 Load Path Connections. All parts of the structure between separation joints shall be interconnected to form a continuous path to the lateral force-resisting system, and the connections shall be capable of transmitting the lateral forces induced by the parts being connected. Any smaller portion of the structure shall be tied to the remainder of the structure with elements having the strength to resist a force of not less than 5% of the portion’s weight.

1.4.2 Lateral Forces. Each structure shall be analyzed for the effects of static lateral forces applied independently in each of two orthogonal directions. In each direction, the static lateral forces at all levels shall be applied simultaneously. For purposes of analysis, the force at each level shall be determined using Eq. (1.4-1):

$$F_x = 0.01W_x \tag{1.4-1}$$

where

F_x = the design lateral force applied at story x , and
 W_x = the portion of the total dead load of the structure, D , located or assigned to level x .

Structures explicitly designed for stability, including second-order effects, shall be deemed to comply with the requirements of this section.

1.4.3 Connection to Supports. A positive connection for resisting a horizontal force acting parallel to the member shall be provided for each beam, girder, or truss either directly to its supporting elements or to slabs designed to act as diaphragms. Where the connection is through a diaphragm, the member’s supporting element shall also be connected to the diaphragm. The connection shall have the strength to resist a force of 5% of the unfactored dead load plus live load reaction imposed by the supported member on the supporting member.

1.4.4 Anchorage of Structural Walls. Walls that provide vertical load bearing or lateral shear resistance for a portion of the structure shall be anchored to the roof and all floors and members that provide lateral support for the wall or that are supported by the wall. The anchorage shall provide a direct connection between the walls and the roof or floor construction. The connections shall be capable of resisting a strength level horizontal force perpendicular to the plane of the wall equal to 0.2 times the weight of the wall tributary to the connection, but not less than 5 psf (0.24 kN/m²).

1.4.5 Extraordinary Loads and Events. When considered, design for resistance to extraordinary loads and events shall be in accordance with the procedures of Section 2.5.

1.5 CLASSIFICATION OF BUILDINGS AND OTHER STRUCTURES

1.5.1 Risk Categorization. Buildings and other structures shall be classified, based on the risk to human life, health, and welfare associated with their damage or failure by nature of their occupancy or use, according to Table 1.5-1 for the purposes of applying flood, wind, snow, earthquake, and ice provisions. Each building or other structure shall be assigned to the highest applicable Risk Category or Categories. Minimum design loads for structures shall incorporate the applicable importance factors given in Table 1.5-2, as required by other sections of this standard. Assignment of a building or other structure to multiple Risk Categories based on the type of load condition being evaluated (e.g., snow or seismic) shall be permitted.

When the building code or other referenced standard specifies an Occupancy Category, the Risk Category shall not be taken as lower than the Occupancy Category specified therein.

Table 1.5-1 Risk Category of Buildings and Other Structures for Flood, Wind, Snow, Earthquake, and Ice Loads

Use or Occupancy of Buildings and Structures	Risk Category
Buildings and other structures that represent low risk to human life in the event of failure	I
All buildings and other structures except those listed in Risk Categories I, III, and IV	II
Buildings and other structures, the failure of which could pose a substantial risk to human life	III
Buildings and other structures, not included in Risk Category IV, with potential to cause a substantial economic impact and/or mass disruption of day-to-day civilian life in the event of failure	
Buildings and other structures not included in Risk Category IV (including, but not limited to, facilities that manufacture, process, handle, store, use, or dispose of such substances as hazardous fuels, hazardous chemicals, hazardous waste, or explosives) containing toxic or explosive substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released ^a	
Buildings and other structures designated as essential facilities	IV
Buildings and other structures, the failure of which could pose a substantial hazard to the community	
Buildings and other structures (including, but not limited to, facilities that manufacture, process, handle, store, use, or dispose of such substances as hazardous fuels, hazardous chemicals, or hazardous waste) containing sufficient quantities of highly toxic substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released ^a	
Buildings and other structures required to maintain the functionality of other Risk Category IV structures	

^aBuildings and other structures containing toxic, highly toxic, or explosive substances shall be eligible for classification to a lower Risk Category if it can be demonstrated to the satisfaction of the Authority Having Jurisdiction by a hazard assessment as described in Section 1.5.3 that a release of the substances is commensurate with the risk associated with that Risk Category.

Table 1.5-2 Importance Factors by Risk Category of Buildings and Other Structures for Snow, Ice, and Earthquake Loads

Risk Category from Table 1.5-1	Snow Importance Factor, I_s	Ice Importance Factor—Thickness, I_t	Ice Importance Factor—Wind, I_w	Seismic Importance Factor, I_e
I	0.80	0.80	1.00	1.00
II	1.00	1.00	1.00	1.00
III	1.10	1.15	1.00	1.25
IV	1.20	1.25	1.00	1.50

Note: The component importance factor, I_p , applicable to earthquake loads, is not included in this table because it depends on the importance of the individual component rather than that of the building as a whole, or its occupancy. Refer to Section 13.1.3.

1.5.2 Multiple Risk Categories. Where buildings or other structures are divided into portions with independent structural systems, the classification for each portion shall be permitted to be determined independently. Where building systems, such as required egress, HVAC, or electrical power, for a portion with a higher Risk Category pass through or depend on other portions of the building or other structure having a lower Risk Category, those portions shall be assigned to the higher Risk Category.

1.5.3 Toxic, Highly Toxic, and Explosive Substances. Buildings and other structures containing toxic, highly toxic, or explosive substances are permitted to be classified as Risk Category II structures if it can be demonstrated to the satisfaction of the Authority Having Jurisdiction by a hazard assessment as part of an overall risk management plan (RMP) that a release of the toxic, highly toxic, or explosive substances is not sufficient to pose a threat to the public.

To qualify for this reduced classification, the owner or operator of the buildings or other structures containing the toxic, highly toxic, or explosive substances shall have an RMP that incorporates three elements as a minimum: a hazard assessment, a prevention program, and an emergency response plan.

As a minimum, the hazard assessment shall include the preparation and reporting of worst-case release scenarios for each structure under consideration, showing the potential effect on the public for each. As a minimum, the worst-case event shall include the complete failure e.g., instantaneous release of entire contents of a vessel, piping system, or other storage structure. A worst-case event includes, but is not limited to, a release during the design wind or design seismic event. In this assessment, the evaluation of the effectiveness of subsequent measures for accident mitigation shall be based on the assumption that the complete failure of the primary storage structure has occurred. The off-site impact shall be defined in terms of population within the potentially affected area. To qualify for the reduced classification, the hazard assessment shall demonstrate that a release of the toxic, highly toxic, or explosive substances from a worst-case event does not pose a threat to the public outside the property boundary of the facility.

As a minimum, the prevention program shall consist of the comprehensive elements of process safety management, which is based upon accident prevention through the application of management controls in the key areas of design, construction, operation, and maintenance. Secondary containment of the toxic, highly toxic, or explosive substances; including, but not limited to, double-wall tank, dike of sufficient size to contain a spill, or other means to contain a release of the toxic, highly toxic, or explosive substances within the property boundary of the facility and prevent release of harmful quantities of contaminants to the air, soil, groundwater, or surface water; are permitted to be used to mitigate the risk of release. Where secondary containment is provided, it shall be designed for all environmental loads and is not eligible for this reduced classification. In hurricane-prone regions, mandatory practices and procedures that effectively diminish the effects of wind on critical structural elements or that alternatively protect against harmful releases during and after hurricanes are permitted to be used to mitigate the risk of release.

As a minimum, the emergency response plan shall address public notification, emergency medical treatment for accidental exposure to humans, and procedures for emergency response to releases that have consequences beyond the property boundary of the facility. The emergency response plan shall address the potential that resources for response could be compromised by the event that has caused the emergency.

1.6 ADDITIONS AND ALTERATIONS TO EXISTING STRUCTURES

When an existing building or other structure is enlarged or otherwise altered, structural members affected shall be strengthened if necessary so that the factored loads defined in this document will be supported without exceeding the specified design strength for the materials of construction. When using allowable stress design, strengthening is required when the stresses caused by nominal loads exceed the specified allowable stresses for the materials of construction.

1.7 LOAD TESTS

A load test of any construction shall be conducted when required by the Authority Having Jurisdiction whenever there is reason to question its safety for the intended use.

1.8 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

OSHA Standards for General Industry, 29 CFR (Code of Federal Regulations) Part 1910.1200, Appendix A, with Amendments as of February 1, 2000, U.S. Department of Labor, Occupational Safety and Health Administration, 2005.

Cited in: Section 1.2

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